



ESG peer spillover effect in supply chain: Evidence from Taiwan semiconductor industry

Donald Lien^a, Lie-Huey Wang^{b,*}

^a Alvarez College of Business, The University of Texas at San Antonio, USA

^b Department of Finance, Ming Chuan University, 250 Zhong-Shan N. Rd., Sec., 5, 111, Taipei, Taiwan, ROC

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ABSTRACT

In this study, we collected data from 271 semiconductor companies listed on the Taiwan Stock Exchange from 2015 to 2020 to determine whether environmental, social, and governance (ESG) spillovers propagate along supply chains through horizontal peers and vertical upstream, midstream, and downstream sectors. The results indicate the presence of a horizontal peer spillover effect that was more significant in the upstream industries than in the midstream and downstream industries. In addition, the forward industry (customers) had a positive vertical spillover effect on the backward industry (suppliers). We also identified a significant moderating effect from high-revenue linkages but not from high-cost linkages. Taken together, these findings indicate that through network linkages, Taiwan's semiconductor supply chain has a vertical spillover effect on the environmental pillar of ESG scores.

1. Introduction

Environmental, social, and governance (ESG) indicators play a key role in the world economy and are regarded as crucial concerns by governments, companies, and investors. Companies often face major external pressure to integrate ESG objectives into their business models (Benabou & Tirole, 2010). ESG indicators influence companies' profitability (Brogi & Lagasio, 2018), stock returns and liquidity (Bolton & Kacperczyk, 2021; de Souza Cunha et al., 2020; Dai et al., 2021; Feng et al., 2022; Luo, 2022; Pedersen et al., 2021; Roy et al., 2022; Welch & Yoon, 2023), financing decisions (Limkriangkrai et al., 2017), bank lending (Houston & Shan, 2022), and consumer satisfaction and brand image (Koh et al., 2022). In addition, companies' engagement in ESG activities can mitigate their downside risk (Hoepner et al., 2024), and studies have reported an association between ESG ratings and loan pricing (Goss & Roberts, 2011; Hasan et al., 2017). Given the major effect of ESG indicators on companies' financial and market performance (Ahmad et al., 2021; Zhao et al., 2018), investors and analysts tend to consider these indicators as a primary component in analyzing the market value and financial performance of companies (Camoletto et al., 2022; Darnall et al., 2022). Therefore, companies have increasingly disclosed their ESG data in response to growing demands by investors for information that can enable them to understand how the companies navigate problems that may affect their performance.

Companies with high ESG performance tend to gain more market advantages and business opportunities (Ahmad et al., 2021; Brogi & Lagasio, 2018; Koh et al., 2022; Zhao et al., 2018). By contrast, companies that do not consider ESG sustainability face the risk of being eliminated from the market. That is, ESG indicators are factors that represent the competitiveness of companies and an entry

* Corresponding author.

E-mail addresses: Don.Lien@utsa.edu (D. Lien), lhwang@mail.mcu.edu.tw (L.-H. Wang).

threshold for obtaining customer orders (Costello, 2013; Crawford et al., 2020; Darendeli et al., 2022). Failure to meet ESG standards established within a supply chain may cause a loss in customer orders (Dai et al., 2021; She, 2022). For example, in response to consumer pressure, South Korea's Samsung, one of the world's top three semiconductor manufacturers, has announced that it is preparing to join RE100,¹ an international organization that advocates for the use of green electricity.² In Taiwan, Delta Electronics once faced a risk of losing orders because it failed to provide an RE100 certificate, which its major customers expected from it, in time.³ Taiwan Semiconductor Manufacturing Company (TSMC) listed the carbon footprint and carbon reduction performance of its supply chain as key indicators in its purchasing policy and announced that it expected its suppliers to reduce their energy use by 20 % by 2030, with this announcement affecting more than 700 suppliers.⁴ Hon Hai Precision Industry attracted consumer attention because of its carbon emissions and labor problems. It has incorporated ESG indicators into its policy to reverse its negative image with respect to personnel management.⁵

Sustainable supply chains are a key component in the sustainable development of international enterprises. Previously, supplier assessments were focused on quality, cost, delivery, safety, and productivity requirements. However, in the modern age, because ESG sustainable development has become a concern, sustainability-related concerns such as the environment (E), social responsibility (S), and corporate governance (G) must be considered in assessments. In consideration of this, companies have increasingly begun to collaborate to modify their supply chains and encourage more collective ESG action. According to the literature, enterprises must collaborate to achieve sustainable development by leveraging the collective power of members of supply chains (Cherrafi et al., 2022; Correia et al., 2024; Cui et al., 2024; Govindan et al., 2021; Vachon & Klassen, 2008; Veldman et al., 2023). The concept of sustainable supply chain management (SSCM) has emerged in response to growing public concern regarding environmental and social problems (Balon, 2020; Khan et al., 2020; Koberg & Longoni, 2019). In this context, buyer-supplier collaboration facilitates improvements in economic, social, and environmental performance. In buyer-supplier relationships, information sharing has a major effect on decision-making processes (Zhu et al., 2021) and thus has a direct influence on supply chain performance (Ada et al., 2021; Kankam et al., 2023), and empirical evidence indicates that corporate social responsibility (CSR) is a key component in the customer-supplier relationship (Dai et al., 2021; Darendeli et al., 2022; She, 2022).

Taiwan's semiconductor industry has the most complete industry cluster and division of specialized labor in the world (Appendix A1).⁶ This ecosystem cannot be replicated or replaced by other countries. Because of this, Taiwan's semiconductor industry plays a pivotal role in both the Taiwanese and the global supply chain. Starting in 2000, the growth rate of Taiwan's semiconductor output has been three times higher than the global average, with its advanced processes below 10 nm accounting for 69 % of the global supply and its mature process production capacity being the highest in the world.⁷ Taiwan ranks higher than South Korea, Japan, Europe, China, and Singapore as a supplier of semiconductors. Taiwan's TSMC and United Microelectronics (UMC) rank first and third, respectively, worldwide as global foundry leaders, with these organizations having market shares of 52.1 % and 7 %, respectively.⁸

Although its development of semiconductors has increased Taiwan's international competitiveness and economic benefits, it has also caused environmental safety problems, including the excessive utilization of hydropower resources and the release of greenhouse gases and waste. As TSMC's role in the global semiconductor industry has become more pivotal, its burden on the environment has increased. The development of artificial intelligence and the predicted exacerbation of environmental protection challenges have rendered consideration of environmental sustainability and meeting ESG requirements crucial to the future development of Taiwan's semiconductor industry.

Sustainability, ESG performance, and carbon reduction are novel considerations in the semiconductor industry. Nevertheless, companies often face pressure from peers in their supply chains to meet ESG requirements, and therefore, they must improve their ESG performance to survive future competition. Companies within global semiconductor industries have been driven to adopt sustainable frameworks because of industry norms and because of a need for such frameworks for interaction within supply chains. Notably, the Semiconductor Industry Association has established the Semiconductor Climate Consortium to enhance the disclosure of carbon emission data by international semiconductor companies.

In Taiwan, leading semiconductor companies often collaborate with customers and suppliers to achieve SSCM. For example, TSMC has the world's first industrial recycled water plant and zero-waste manufacturing center for recycling wastewater and waste. It has also helped suppliers establish an environmental protection system to reduce emissions and encouraged them to employ carbon-capturing equipment. UMC has invested in research and development in the field of integrated circuit (IC) design for green products and has established the Triple R Alliance to encourage suppliers to engage in recycling and energy saving. MediaTek has

¹ RE100 refers to 100 % renewable energy, as proposed by Climate Group and the Carbon Disclosure Project. Members of RE100 must publicly make a commitment to achieving the goal of using 100 % renewable energy by 2050.

² Energy Tracker Asia Newsletter (2022, <https://energytracker.asia/samsung-joins-re100/>).

³ <https://www.deltaww.com/en-us/news/14986>.

⁴ 2023 Sustainability Report of TSMC.

⁵ 2021 Sustainability Report of Foxconn Technology Group.

⁶ Wafer manufacturing is the core of Taiwan's semiconductor industry. The upstream involves integrated circuit (IC)/silicon intellectual property (SIP) design, covering 16 product categories, with a self-contained system that extends to downstream packaging and testing. The midstream involves IC/wafer manufacturing, covering six product categories. The downstream involves IC testing and packaging, covering six product categories. Appendix A1 lists the 28 categories covered in Taiwan's semiconductor supply chain.

⁷ Frontline Formosa (2023). *The Economist*, special report.

⁸ 2023 research report from TrendForce.

implemented green innovation strategies in the node design of advanced processing technologies to reduce chip power consumption and size. Novatek Microelectronics has engaged in IC process miniaturization to reduce the power consumption and carbon emissions of its end products. Furthermore, ASE Technology Holding has adopted a green building concept and green factory layouts.

The majority of studies in ESG field have focused on customer sustainable responsibility without considering ESG spillovers across peers in the supply chain despite the potential for environmental or social risk at any step along (Dai et al., 2021; Huang et al., 2021; Lian et al., 2023; Schiller, 2018; Tang et al., 2023; Tao et al., 2022). Taiwan's semiconductor industry plays a key in the global market, and its efforts toward improving ESG performance can serve as an example and reference for other countries. Within the unique ecosystem of horizontal and vertical integration in Taiwan's semiconductor supply chain (e.g., central satellite factory systems), companies are able to form network linkages that enable them to collaborate and form alliances. These network linkages affect the companies' financial decisions (Kuo, 2006), and the financial spillover effects observed in Taiwanese supply chains reveal interactive relationships between suppliers and customers, suggesting the coexistence of competitive and cooperative correlations in supplier–customer relationships (Kim & Henderson, 2015). This indicates that ESG peer spillovers may have long-term effects.

In the current study, we analyzed Taiwan's semiconductor industry to determine whether and how ESG spillovers propagate along supply chains through horizontal peers and vertical upstream, midstream, and downstream sectors. We divided peer spillovers into two types: peer spillovers in a specific industry stream, such as an upstream, midstream, or downstream sector (that is, horizontal spillovers), and peer spillovers across upstream, midstream, and downstream sectors (that is, vertical spillovers). We focused on three topics. First, we examined ESG spillovers with a focus on horizontal peers in supply chains. Considering that peers in the same industry stream are regarded as competitors, we hypothesized that the ESG performance of peers would have horizontal spillover effects on companies. Second, we examined ESG spillovers through vertical upstream, midstream, and downstream sectors. Considering that these sectors of a supply chain form customer–supplier relationships, we hypothesized that forward customers and backward suppliers in a supply chain would have vertical peer spillover effects on companies. Third, we explored whether a company's network linkages moderate peer spillovers. Considering that any increase in the degree of network linkage is associated with an increase in competitive pressure and cooperation incentives from horizontal and vertical peers, we hypothesized that network linkages would strengthen peer spillover effects.

This study provides valuable contributions to the literature on sustainable finance. The majority of studies in this area have focused on the effects of customers on the ESG performance of companies, and research on ESG spillovers across peers in supply chains remains limited. In this study, we extended on the literature on the economic implications of ESG practices by companies by introducing a novel perspective through which we demonstrated how spillovers propagate along supply chains. To the best of our knowledge, this is one of the few studies to examine how ESG spillovers propagate along supply chains through horizontal and vertical peers through our consideration of a unique sample of supply chains (Cao et al., 2023; Dai et al., 2021; Lian et al., 2023; Tang et al., 2023).⁹ We observed that horizontal and vertical ESG peer spillovers were present in the supply chain, with horizontal spillovers being more pronounced in the upstream sector. Many studies have used an exclusive-of-the-focal-firm approach to measure peer average scores. In this study, we calculated each focal company's score ranking in subsectors classified on the basis of the Sustainability Accounting Standards Board (SASB), scaled by total rated companies, which is a variable ignored by previous studies. We also provided novel evidence indicating that score ranking plays a key role in horizontal spillovers. We hypothesized that supplier–customer network linkages would have a moderating effect on ESG peer spillovers in supply chains; this topic has rarely been discussed in the literature. Because actual network linkages are difficult to observe or measure, we measured the transaction linkages of customers and suppliers through the net revenue and operating costs of a company's statement of comprehensive income. The results of this study indicate that the net revenue and operating cost network linkages from the upstream, midstream, and downstream sectors of the supply chain had a moderating effect on peer spillovers.

The remainder of this paper is organized as follows. In Section 2, we present a review of the literature on the spillover effects of peers in supply chains and outline our hypotheses. In Section 3, we describe our data, variables, and model specifications. In Section 4, we present and discuss our findings. Finally, in Section 5, we present our conclusions and discuss the implications of our findings.

2. Literature review and hypotheses development

According to conventional wisdom, a company's sole objective is to maximize shareholder interests (Friedman, 1970). Instrumental stakeholder theory posits that companies gain competitive advantages and achieve positive outcomes from engaging in stakeholder management and CSR activities (Donaldson & Preston, 1995; Jones, 1996). Studies indicate that customers tend to actively engage in sustainable management of their suppliers and encourage their suppliers to adopt socially responsible practices. Therefore, when customers adopt CSR policies, suppliers tend to engage more in CSR activities (Dai et al., 2021; Schiller, 2018).

According to the benchmarking mechanism, customers often reevaluate their contracting decisions on the basis of their suppliers'

⁹ In Taiwan, company annual reports must disclose the names of suppliers (customers) that account for more than 10 % of the total purchases (sales) in the past 2 years, as well as the amounts and proportions of their purchases. If the supplier's (customer's) name or transaction partner cannot be disclosed due to contractual agreements, a code name may be used instead. The company's direct customer–supplier information is not readily available, therefore, we are unable to use these data for supplementary analysis. This is a limitation of our research. Although our research could not investigate whether a company's ESG is directly affected by its specific suppliers (customers), we expanded the scope of our research to include peers in the same industry, as well as backward suppliers and forward customers in the supply chain, which includes some of the specific parties that transact with the company.

incremental CSR information (Costello, 2013; Crawford et al., 2020). In addition, according to the mechanism of public pressure, customers often act in response to public pressure to maintain a green or socially responsible supply chain as a means of avoiding reputational damage; protests from customers, employees, and investors; and policymaker pressure (Darendeli et al., 2022; Dimson et al., 2015; Dyreng et al., 2016; Rauter, 2020). Therefore, the quality of supplier CSR generally signals legal and reputational risk exposure and potential supply chain disruptions. According to Dai et al. (2021) and She (2022), customers care not only about their own CSR but also about their suppliers' CSR. Therefore, suppliers' CSR policies can be affected by their customers' CSR practices (Dai et al., 2021). Darendeli et al. (2022) examined the effect of CSR information on noninvestor stakeholder decision-making and supply chain contracting. They identified a negative effect of CSR information shock on suppliers with comparatively low CSR ratings. Generally, such shocks cause these suppliers to experience reductions in the numbers of their contracts and customers.

Studies on SSCM have suggested that suppliers choose to invest in CSR to gain competitive advantages while making positive contributions to society. Huang et al. (2021) argued that SSCM and environmental pressure from stakeholders have a positive effect on the proactiveness of suppliers' environmental strategies, which in turn improves their operational performance but does not improve their environmental performance. Agliardi et al. (2023) asserted that companies with low environmental ratings typically have high financial performance, with environmental leaders having low risks and high resilience. After the Paris Agreement, positive spillovers in clusters with high environmental ratings occurred. In addition, Tao et al. (2022) contended that customers prefer socially responsible suppliers and that this effect becomes more prominent when the supplier industry is more competitive, when customers' CSR performance increases, and when the supplier and customer have a similar focus on CSR. On the basis of these findings, we proposed the following hypotheses:

H1. ESG is subject to a positive horizontal peer spillover effect.

H2. ESG is subject to a positive vertical peer spillover effect.

Interdependencies in supply chain networks can occur across vertical and horizontal boundaries. That is, a supply chain forming economic linkages among a company and its customers, distributors, suppliers, and competitors can affect a company's competitive behavior in the market (Bae et al., 2011). Kuo (2006) argued that network linkages increase company value and reduce reliance on debt financing in Taiwan's electronic industry. Within Taiwan's unique ecosystem, companies tend to rely on horizontal or vertical integration or supply chain relationships to form network linkages and achieve research and development collaboration, information exchange, and industry alliances. When the number of such network linkages increase, the degree of competitive pressure from peers also increases, which in turn influences the horizontal and vertical spillover effects of peers on ESG performance. Considering this, we proposed the following hypothesis:

H3. Network linkages moderate peer spillover effects.

Fig. 1 illustrates our conceptual framework and hypotheses.

3. Data and research design

3.1. Data

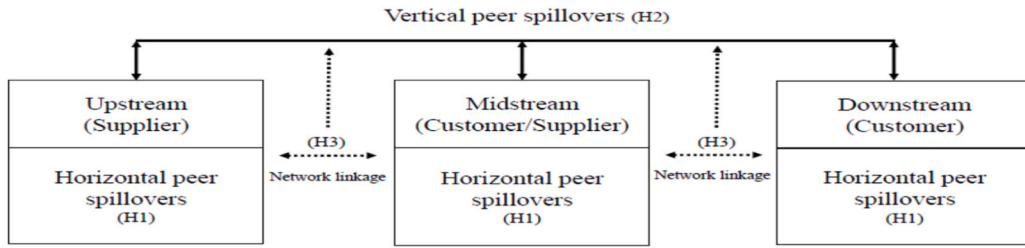
In this study, we empirically analyzed a sample of semiconductor supply chain companies listed on the Taiwan Stock Exchange. This sample comprised 271 semiconductor companies, that is, 101 upstream companies, 44 midstream companies, and 126 downstream companies, with a total observation of 1,571 year-data.¹⁰ We considered the ESG combined score (ESG) and three pillar scores, namely environmental (*Env*), social (*Soc*), and governance (*Gov*) scores, which were obtained from the Taiwan Economic Journal's (TEJ) TSEG Sustainability Indicators database for the period of 2015–2020, as key variables. We also retrieved each company's financial information from the TEJ Finance database.

Each international ESG scoring (or rating) agency has its own framework, and scoring results typically depend on the selected indicators and weights. Consequently, a rated company may receive substantially different ESG ratings from different providers, with unexplained shortcomings (Berg et al., 2022; Chatterji et al., 2016).¹¹ The TSEG factors provided by the TEJ for Taiwanese companies are unique. These factors are divided into 3 pillars and 16 indicators¹² in accordance with the SASB framework. The weights of *Env*-pillar and *Soc*-pillar are those specified by the SASB, whereas the weight of *Gov*-pillar is determined by the TEJ. *Env*-pillar covers greenhouse gas emissions, energy, water and wastewater, waste and toxic substances, and ecological effects. *Soc*-pillar covers human rights and community relations, data security, product quality and safety, employee statistics, employee health and safety, and

¹⁰ Companies with at least 3 years of data were included in the research sample.

¹¹ Berg et al. (2022) decomposed the divergence of ratings from six ESG rating agencies—Kinder, Lydenberg, and Domini; Sustainalytics; Moody's ESG Solutions (Vigeo Eiris); S&P Global (RobecoSAM); Refinitiv (ASSET4); and MSCI—into contributions of scope, measurement, and weight. They reported that divergence, scope, and weights respectively accounted for 56 %, 38 %, and 6 % of the differences in their measurements. They also indicated that the rater effect explained 15 % of the variation observed in category scores. This divergence in ratings poses a challenge for empirical research because utilizing ratings from one rater rather than another may alter a study's results and conclusions.

¹² According to the SASB framework, ESG performance can be divided into 5 categories—environment, social capital, human capital, business model and degree of innovation, and leadership and corporate governance—covering 13 indicators. TEJ's TSEG covers 16 indicators, including the 13 SASB indicators and 3 *Gov*-pillar (control seat, fair stakeholder treatment, and corporate governance information transparency) indicators.



Note: H1 posits that ESG is subject to a positive horizontal peer spillover effect. H2 posits that ESG is subject to a positive vertical peer spillover effect. H3 posits that network linkages moderate peer spillover effects.

Fig. 1. Conceptual framework and hypotheses.

employee diversity. *Gov*-pillar covers business models and innovations, management leadership, control seat, fair stakeholder treatment, and corporate governance information transparency.

TEJ's TSEG scores are calculated using relative rankings within a specific industry (11 major industries of the SASB). All listed, over-the-counter (OTC), emerging, and unlisted public offering companies in Taiwan (approximately 2,300 companies) are rated. The raw ESG quantitative score is the weighted score of quantitative indicators and the Global Reporting Initiative information disclosure of three pillars, with these scores having weights of 0.75 and 0.25, respectively. The scores of *Env*-pillar, *Soc*-pillar, and *Gov*-pillar range between 0 and 100,¹³ with higher scores indicating higher performance with respect to ESG-related indicators.

TESG data are unique for three reasons. First, they are obtained by applying the SASB framework with consideration of negative controversies (e.g., negative ESG news) and adding 3 *Gov*-pillar indicators (control seat, fair stakeholder treatment, and corporate governance information transparency). Second, all listed, OTC, emerging, and unlisted public offering companies are considered in the ratings. Third, ESG scores are calculated using relative rankings within a specific industry, and each pillar is assigned a unique industry weight.

Table 1 presents the distribution of the sample used in the current study by year. Observations of upstream and downstream sectors accounted for 36.41 % and 46.91 % of all observations, respectively. Although the midstream sample accounted for only 16.68 % of all observations, its total assets and net revenue were higher than those of the upstream and downstream samples.

3.2. Variables and empirical model

In this study, we tested horizontal peer spillovers by using the following model:

$$Score_{it} = \alpha_0 + \beta_1 Peer_{jt-1} + \beta_2 Rank_{qt-1} + \sum \delta_k Control_{kt-1} + \theta_t + \mu_i + \varepsilon_{it} \quad (1)$$

where $Score_{it}$ denotes company i 's natural logarithm of ESG, *Env*, *Soc*, and *Gov* scores in year t . These values were obtained from the TEJ's TSEG Sustainability Indicators database for the period of 2015–2020. Two proxy variables were used for peer horizontal spillovers. The first proxy variable was the natural logarithm of a peer average score in a specific upstream, midstream, or downstream sector, excluding the focal company (Drange & Nath, 2019; Li et al., 2023; Yuan et al., 2022). The following equation was used: $Peer_{jt} = \frac{\sum_{q \in j, q \neq i} Score_{qt}}{N_{jt}-1}$, where i denotes the focal company; j denotes the upstream, midstream, or downstream sector¹⁴; q denotes the peer company in industry stream j ; and N_{jt} denotes the total number of companies in industry stream j in year t . Here, $Peer_{jt-1}$ denotes industry stream j 's natural logarithm of the average score of peers in year $t-1$. The second proxy variable was a company's score ranking in subsectors classified by the SASB.¹⁵ The company with the highest score in a specific subsector was ranked as number 1, with the remaining companies also assigned an ordinal ranking. Because the rated companies were assigned different rankings in different SASB subsectors, each company's score ranking was scaled by the total number of rated companies in its SASB subsector. The following equation was used: $Rank_{qt} = 1 - \frac{score_rank_{iq,t}}{N_{qt}}$, where $rank_{qt}$ denotes company i 's score ranking scaled by the total number of rated companies in its subsector q in year t , $score_rank_{iq,t}$ is the score ranking of company i in its SABA subsector q in year t . N_{qt} is the total number of rated companies in subsector q in year t . A higher $rank_{qt}$ value indicates the presence of few peers with higher scores than those of company i in subindustry q . Here, $Rank_{qt-1}$ denotes company i 's score ranking in its SABA subsector q in year $t-1$.¹⁶

¹³ The weighted score of each pillar is multiplied by the industry weight for the pillar, and this score is used as the final score for that pillar.

¹⁴ Industry Value Chain Information Platform (<https://ic.tpex.org.tw/introduce.php?ic=D000>).

¹⁵ The industries within Taiwan's semiconductor supply chain correspond to 6 industries and 13 subsectors in the SASB classification system. Details are provided in Appendix A2.

¹⁶ *Peer* is the natural logarithm of a peer average score calculated by excluding the target company at the same upstream, midstream, or downstream location in the semiconductor industry supply chain. *Rank* denotes a company's score ranking in its SASB subsector. Peers in an SASB subindustry are not the same as peers in a semiconductor industry supply chain (Appendix A2).

Table 1
Sample distribution by year.

	2015	2016	2017	2018	2019	2020	Total
Upstream	88	89	92	101	101	101	572 (36.41 %)
Midstream	43	43	44	44	44	44	262 (16.68 %)
Downstream	118	119	123	126	126	125	737 (46.91 %)
Total	249	251	259	271	271	270	1571 (100 %)

Note: This table presents the distribution of observations in the upstream, midstream, and downstream sectors of Taiwan's semiconductor supply chain from 2015 to 2020.

As previously performed in the literature (Behl et al., 2022; Dai et al., 2021; Li et al., 2023; Tang et al., 2023), we included an array of lagged company-level characteristics as control variables (*Control*). These characteristics included company size, measured as the natural logarithm of total assets (*Lnta*); profitability, measured as the ratio of earnings before interest and tax (EBIT) to average total assets (*ROA*); leverage, measured as the ratio of total debt to total assets (*DebtRatio*); and growth, measured as the ratio of the market value of equity to the book value of equity (*MBratio*). Here, θ_t is the year fixed effect (FE),¹⁷ μ_i is the firm fixed effect (FE), and ε_{it} is the error term.

We tested vertical peer spillovers by using the following model:

$$Score_{it} = \alpha_0 + \sum \beta_j (Up|Mid|Down)_{jt-1} + \sum \delta_k Control_{kt-1} + \theta_t + \mu_i + \varepsilon_{it} \quad (2)$$

where Up_{jt-1} , Mid_{jt-1} , and $Down_{jt-1}$ denote the natural logarithms of the average scores of industry stream j in the upstream, midstream, and downstream sectors in year $t-1$, respectively. The natural logarithms of the peer average scores of the upstream, midstream, and downstream sectors were used as proxy variables for vertical peer spillovers. The following equation was used: $Up|Mid|Down_{jt} = \frac{\sum_{q \in j} Score_{qt}}{N_{jt}}$, where $Up|Mid|Down_{jt}$ is the average scores of upstream, midstream, or downstream j in year t , $Score_{qt}$ is the score of peer company q in industry stream j in year t , and N_{jt} is the total number of companies in industry stream j in year t . The other variables are the same as those in Eq. (1).

After including the interaction terms of network linkages into models (1) and (2), we tested the moderating effect on peer spillovers:

$$Score_{it} = \alpha_0 + \beta_1 Peer_{jt-1} + \beta_2 Rank_{qt-1} + \beta_3 Peer_{jt-1} \times Linkage_{it-1} + \beta_4 Rank_{qt-1} \times Linkage_{it-1} + \sum \delta_k Control_{kt-1} + \theta_t + \mu_i + \varepsilon_{it} \quad (3)$$

$$Score_{it} = \alpha_0 + \sum \beta_j (Up|Mid|Down)_{jt-1} + \sum \gamma_j (Up|Mid|Down)_{jt-1} \times Linkage_{it-1} + \sum \delta_k Control_{kt-1} + \theta_t + \mu_i + \varepsilon_{it} \quad (4)$$

where $Linkage_{it-1}$ represents company i 's network linkage in year $t-1$. The other variables were the same as those in Eqs. (1) and (2).

Observing the direct or actual network linkages between companies and their customers and suppliers (company-to-company transaction) is difficult. Therefore, Kuo (2006) employed productive and financial network linkages as proxy variables.¹⁸ Specifically, he uses the number of original equipment manufacturer (OEM) or original design manufacturer (ODM) satellite companies as a proxy for productive linkage and the ratio of net working capital to net revenue as a proxy for financial linkage. Typically, the degree of customer-supplier linkage involved in a transaction can be indirectly determined from the net revenue and operating costs outlined by a company's statement of comprehensive income. When the net revenue increases, the degree of linkage between a company and its customers also increases. Additionally, when the operating costs increase, the degree of linkage between a company and its suppliers increases. This suggests that more net revenue and operating cost linkages reflect greater competition and cooperation pressure from peers. Therefore, the transaction linkage between customers and suppliers could affect spillovers. In this study, with reference to previous studies on strategic network linkages (Jarillo, 1988; Kuo, 2006), we adopted two proxy variables to measure the degree of supplier-customer transaction linkages in a supply chain rather than considering actual linkages. In other words, we measured net revenue by using the natural logarithm of net revenue, and we measured operating costs by considering the ratio of operating cost to net revenue (that is, operating cost ratio). After sorting the net revenue (operating cost ratio) of all companies in descending order by year, we divided the sample into three equal groups. We established two dummy variables *highsale* and *highcost* that were assigned a value of 1 when company's net revenue and operating cost ratio is the top 1/3 of all companies and a value of 0 otherwise.¹⁹ These two dummy variables were used to represent network linkages between companies and their customers and suppliers.

Appendix A3 provides a detailed description of all variables. Equations (1)–(4) were estimated using panel data regression with

¹⁷ A year fixed effect was used to control for the effect of annual economic fluctuations on a company's score ranking in subsectors.

¹⁸ Kuo (2006) measured productive network linkages by using the number of OEM/ODM satellite companies (intercompany organizations) and measured financial network linkages by using the ratio of net working capital to net revenue.

¹⁹ The sample grouping was not clearly defined. Most studies rely on medians to differentiate between high and low groups; however, this approach may cause observations close to the median to be poorly distinguishable. With consideration of the nature of panel data, net revenue and operating cost ratio groupings are typically determined on a year-by-year basis. In this study, we divided the sample into five or six groups and achieved qualitatively similar results.

robust standard errors clustered at the company level (White, 1980). To reduce the occurrence of outliers, all continuous variables can be winsorized at the 1st and 99th percentiles, as outlined by Galletta et al. (2021), Galletta and Mazzu (2022), Khan et al. (2017), and Kohler (2015). However, because ESG scores are subjected to several tests and control procedures by the TEJ, we did not apply winsorization.

4. Results and discussion

4.1. Descriptive statistics and correlation analysis

Table 2 presents the descriptive statistics of all variables. The ESG means were 54.64, 57.14, and 54.03 for the upstream, midstream, and downstream sectors, respectively. The Env-pillar means were 52.43, 58.39, and 53.07 for the upstream, midstream, and downstream sectors, respectively. The Soc-pillar means were 56.65, 58.79, and 54.37 for the upstream, midstream, and downstream sectors, respectively. These results indicate that the midstream sector outperformed the other two stream sectors in terms of ESG and the three pillar scores.

As listed in Table 2, the total assets were NT\$11771 million, NT\$78267 million, and NT\$18601 million for the upstream, midstream, and downstream sectors, respectively. In addition, the net revenues were NT\$8067 million, NT\$41468 million, and NT\$29877 million for the upstream, midstream, and downstream sectors, respectively. In other words, the average total assets of the midstream sector were 6.6 and 4.2 times those of the upstream and downstream sectors, respectively. Additionally, the average net revenues of the midstream sector were 5.1 and 1.4 times those of the upstream and downstream sectors, respectively. The distribution of total assets, net income and operating costs of the upstream, midstream and downstream sectors present a right skewness.

As presented in Panel D of Table 2, the results of Wald's chi-square test of the means of the variables in the upstream, midstream, and downstream sectors revealed significant differences in ESG, Env-pillar, and Soc-pillar, but not in Gov-pillar. Since 2003, the Taiwanese government has proposed several corporate governance policies applicable to all listed companies. For example, the Financial Supervisory Commission released the Strengthening Corporate Governance Blueprint in 2013 and the New Corporate Governance Blueprint in 2018. In 2020, it released the Corporate Governance 3.0 Sustainable Development Blueprint, which incorporated environmental and sustainability factors and was proposed to encourage companies to voluntarily invest in ESG sustainable development. It also required listed companies in Taiwan to consider three indicators in Gov-pillar (i.e., control seat, fair stakeholder treatment, and corporate governance information transparency). Listed companies that violate the aforementioned regulations are typically punished, which may explain the current findings of a lack of a significant difference in the Gov-pillar scores across the three industry stream sectors.

Table 3 depicts a Pearson's correlation analysis matrix of the variables used in the regression model. As indicated by the correlation matrix for horizontal spillovers in Panel A, the correlation coefficients between the ESG scores and peers' average scores ranged from -0.0249 to 0.0525 , the correlation coefficients between the ESG scores and score rankings ranged from 0.2357 to 0.9677 , and the correlation coefficients between score rankings and peers' average scores ranged from -0.303 to 0.049 . Notably, the TEJ's TESG score is a weighted average of three pillar scores, with each pillar score converted from an SASB ranking. Rank is based on a score's ranking in an SASB subsector. Score is likely to exhibit a strong correlation with Rank. Peer is the average score of other companies in the same stream industry. Differences in industry classification may lead to a weak correlation between Peer and Score. In this study, the correlation coefficients observed for the company-level characteristic variables ranged from -0.1458 to 0.5820 . In addition, the correlation coefficient observed between Env-pillar and Soc-pillar was 0.4964 , and the correlation coefficient observed between Env-rank and Soc-rank was 0.4736 . These results indicate that companies with a strong focus on social responsibility focused on environmental protection. As indicated by the correlation matrix for vertical spillovers in Panel B of Table 3, a significantly negative correlation was observed between the upstream (or midstream) and downstream sectors, along with a positive correlation between the midstream and downstream sectors.

We tested the collinearity of the independent variables in the regression model. Table 4 lists the variance inflation factors (VIFs) of these variables. In the vertical spillover model, the VIF values in the upstream sector ranged from 1.16 to 2.64, with an average of 1.72; the VIF values in the midstream sector ranged from 1.16 to 5.36, with an average of 2.62; and the VIF values in the downstream sector ranged from 1.16 to 2.47, with an average of 1.66. Both the VIF values of the individual variables and the average VIF value were less than 10, indicating the absence of a major collinearity problem among the independent variables in the regression model.

4.2. Peer spillovers in supply chains

Table 5 presents the results regarding ESG spillover effects in supply chains. Analysis of horizontal spillovers (Table 5, Panel A) revealed a significantly positive Peer spillover effect on the Env-pillar score in the upstream sector and a significantly negative Peer spillover effect on the Env-pillar score in the downstream sector. However, no significant Peer spillover effects were observed in the midstream. From 2015 to 2020, the average number of companies in the upstream, midstream, and downstream sectors was 95, 44, and 262, respectively (Table 1). Notably, the assets and net revenues of the upstream companies were smaller than those of the midstream companies, with many peer companies existing in the upstream. According to the mechanisms of benchmarking and public pressure, upstream companies typically experience two forms of competitive pressures. The first stems from the possibility of midstream customers reevaluating their contracts on the basis of their suppliers' ESG performance, and the second originates from competition with upstream peers (Costello, 2013; Crawford et al., 2020; Darendeli et al., 2022; Rauter, 2020). Upstream companies tend to closely monitor the average scores of their peers and their score rankings in the SASB subsectors, which may explain the

Table 2
Summary of descriptive statistics.

Panel A. Upstream sector/Variable	Mean	SD	Min	Max	p25	p50	p75
ESG	54.64	6.78	35.44	74.50	49.54	53.88	59.30
Env	52.43	9.53	33.85	84.74	45.21	45.94	60.07
Soc	56.65	6.65	37.97	89.13	52.70	55.95	59.42
Gov	54.02	10.48	22.45	78.18	45.79	53.87	61.81
Total assets (NT\$ million)	11771	46986	80	533906	731	1681	5361
Net revenue (NT\$ million)	8067	28377	13	322146	473	1219	4334
Operating cost (NT\$ million)	5159	17328	0	180610	254	758	2675
ROA	5.33	11.77	−112.51	34.62	0.06	6.22	12.06
Debratio	26.64	14.88	0.49	94.24	16.59	23.94	33.42
MBratio	2.97	3.53	0.43	43.80	1.18	1.94	3.16
Panel B. Midstream sector/Variable	Mean	SD	Min	Max	p25	p50	p75
ESG	57.14	8.89	37.46	80.34	49.83	56.62	63.03
Env	58.39	12.16	32.53	88.61	47.11	55.81	68.09
Soc	58.79	11.02	36.89	91.00	51.13	57.09	65.19
Gov	54.85	11.94	23.06	76.58	46.86	55.93	62.87
Total assets (NT\$ million)	78267	324047	552	2760712	2146	4036	20759
Net revenue (NT\$ million)	41468	160080	36	1339255	1357	3803	11894
Operating cost (NT\$ million)	25569	83310	58	628124	1127	2921	8827
ROA	1.21	8.88	−30.76	31.24	0.89	5.25	9.07
Debratio	38.01	18.75	4.69	93.99	25.67	36.58	47.96
MBratio	2.25	2.81	0.26	41.40	1.13	1.69	2.68
Panel C. Downstream sector/Variable	Mean	SD	Min	Max	p25	p50	p75
ESG	54.03	7.45	35.95	78.41	48.45	52.85	58.42
Env	53.07	9.74	32.60	85.11	45.27	51.40	59.83
Soc	54.37	8.93	36.15	83.49	48.02	53.03	59.62
Gov	54.32	10.83	23.71	78.23	46.53	54.96	61.85
Total assets (NT\$ million)	18601	64775	139	684346	1632	4051	10961
Net revenue (NT\$ million)	29877	132293	61	1399333	1377	3362	9674
Operating cost (NT\$ million)	27621	126412	15	1349729	1068	2568	7805
ROA	4.73	6.38	−38.01	34.11	2.33	5.14	7.84
Debratio	43.12	18.56	1.23	84.08	28.74	43.81	57.37
MBratio	1.61	1.29	0.36	13.48	0.95	1.26	1.77
Panel D. Difference in means for variables in the upstream, midstream, and downstream sectors/ Variable	Upstream	Midstream	Downstream	Wald χ^2			
ESG	54.64	57.14	54.03	25.60***			
Env	52.43	58.39	53.07	50.97***			
Soc	56.65	58.79	54.37	46.99***			
Gov	54.02	54.85	54.32	0.97			
Total assets (NT\$ million)	11771	78267	18601	14.96***			
Net revenue (NT\$ million)	8067	41468	29877	29.36***			
Operating cost (NT\$ million)	5159	25569	27621	37.36***			
ROA	5.33	1.21	4.73	2.40			
Debratio	26.64	38.01	43.12	327.93***			
MBratio	2.97	2.25	1.61	84.91***			

Notes: This table presents the descriptive statistics for the study variables. Panels A, B, and C present the descriptive statistics of the upstream, midstream, and downstream sectors, respectively. Panel D presents the results of Wald's chi-square (Wald χ^2) test for the means of the variables in the upstream, midstream, and downstream sectors. *ESG*, *Env*, *Soc*, and *Gov* denote the *ESG* combined score, *Env*-pillar score, *Soc*-pillar score, and *Gov*-pillar score, respectively. *ROA* denotes the ratio of EBIT to average total assets. *Debt ratio* denotes ratio of total debt to total assets. *Mbratio* denotes the ratio of market value of equity to book value of equity. p25, p50, and p75 indicate the 25th, 50th, and 75th percentiles, respectively.

significant peer spillover effects observed in the *Env*-pillar scores.

In Taiwan, midstream companies, including TSMC, which holds a leading position, are predominantly large entities with extensive assets and revenue, and the spillover effect among peers for these companies is nonsignificant. By contrast, the downstream sector includes numerous peer companies in widely varying SASB subsectors. This diversity is likely one of the reasons for the negative peer spillover effect observed in the *Env*-pillar scores in the present study. We also observed a significantly positive *Rank* spillover effect on the *ESG* combined scores and *Gov*-pillar scores across the upstream, midstream, and downstream sectors. Overall, our findings indicate that the spillover effect of peers' average scores is strong in the upstream sector but weak in the midstream and downstream sectors.

In the current study, in the upstream sector, vertical spillover effects were observed in the *Soc*-pillar scores but not in the *ESG* combined scores, *Env*-pillar scores, or *Gov*-pillar scores (Table 5, Panel B). The *Soc*-pillar scores in the upstream sector were lower than those in the midstream sector but higher than those in the downstream sector (Table 2). These findings indicate that the midstream sector has a positive vertical spillover effect on the upstream sector, whereas the downstream sector has a negative vertical spillover effect.

Table 3
Pearson correlation matrix.

Panel A. Variables for horizontal spillovers

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
(1) <i>InESG</i>	1															
(2) <i>InEnv</i>	0.6816***	1														
(3) <i>InSoc</i>	0.7642***	0.4964***	1													
(4) <i>InGov</i>	0.7762***	0.2468***	0.2953***	1												
(5) <i>ESGpeer</i>	0.0226	0.0288	0.0306	-0.0076	1											
(6) <i>Envpeer</i>	0.0273	0.0030	0.0297	0.0177	0.9079***	1										
(7) <i>Socpeer</i>	0.0291	0.0525***	0.0226	-0.0026	0.5908***	0.7236***	1									
(8) <i>Govpeer</i>	-0.0131	-0.0092	0.0065	-0.0249	0.4839***	0.1828***	-0.4008***	1								
(9) <i>ESGrank</i>	0.9487***	0.6416***	0.7092***	0.7482***	0.0322	0.0402	0.0278	-0.0050	1							
(10) <i>Envrank</i>	0.6626***	0.9677***	0.4893***	0.2357***	0.0220	-0.0008	0.0410	-0.0070	0.6517***	1						
(11) <i>Socrank</i>	0.7046***	0.4736***	0.9484***	0.2414***	0.0470	0.0479*	0.0369	0.0080	0.6959***	0.4819***	1					
(12) <i>Govrank</i>	0.7793***	0.2802***	0.3162***	0.9560***	0.0027	0.0307	0.0117	-0.0303	0.7702***	0.2697***	0.2611***	1				
(13) <i>Lnta</i>	0.5820***	0.4929***	0.5564***	0.3010***	-0.0122	-0.0043	0.0339	-0.0469*	0.5367***	0.4837***	0.5273***	0.3085***	1			
(14) <i>ROA</i>	0.2775***	0.1542***	0.2574***	0.2046***	-0.0683***	-0.0509***	-0.0715***	-0.0110	0.2862***	0.1368***	0.2842***	0.2045***	0.2906***	1		
(15) <i>DebtRatio</i>	0.0285	0.1388***	-0.0190	-0.0139	0.0004	-0.0133	0.0426*	-0.0331	0.0160	0.1113***	-0.0187	-0.0053	0.2879***	-0.1458***	1	
(16) <i>Mbratio</i>	0.0878***	0.0497***	0.0609***	0.0792***	-0.1349***	-0.1375***	-0.1494***	0.0146	0.0917***	0.0387*	0.0759***	0.0758***	-0.0943***	0.3292***	-0.1010***	1

Panel B. Variables for vertical spillovers

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
(1) <i>InESG</i>	1																			
(2) <i>InEnv</i>	0.6816***	1																		
(3) <i>InSoc</i>	0.7642***	0.4964***	1																	
(4) <i>InGov</i>	0.7762***	0.2468***	0.2953***	1																
(5) <i>ESGpeer</i>	0.0150	0.0342	0.0236	-0.0135	1															
(6) <i>ESGmid</i>	-0.0076	-0.0146	-0.0197	0.0118	-0.8082***	1														
(7) <i>ESGdown</i>	-0.0110	-0.0246	-0.0182	0.0114	-0.9333***	0.8052***	1													
(8) <i>EnvGap</i>	0.0161	0.0381	0.0268	-0.0162	0.9459***	-0.7398***	-0.7824***	1												
(9) <i>Envmid</i>	-0.0071	-0.0119	-0.0187	0.0096	-0.7231***	0.9715***	0.6640***	-0.7016***	1											
(10) <i>Envdown</i>	0.0124	0.0378	0.0176	-0.0120	0.2373	0.1525***	0.0271	0.3962***	0.1328***	1										
(11) <i>Socup</i>	0.0137	0.0336	0.0271	-0.0177	0.9499	-0.8784***	-0.8632***	0.9283***	-0.8162***	0.2923***	1									
(12) <i>Socmid</i>	0.0013	0.0082	-0.0007	-0.0028	-0.5019	0.7284***	0.7079***	-0.2507***	0.6478***	0.6187***	-0.4542***	1								
(13) <i>Socdown</i>	-0.0144	-0.0309	-0.0208	-0.0094	-0.9536	0.8107***	0.8958***	-0.8525***	0.7471***	-0.9182***	0.5902***	-0.9182***	1							
(14) <i>Govup</i>	0.0112	0.0202	0.0077	0.0061	0.7883	-0.4927***	-0.8525***	0.6109***	-0.3659***	-0.0573***	-0.6240***	-0.7816***	-0.6240***	1						
(15) <i>Govmid</i>	-0.0114	-0.0276	-0.0208	0.0181	-0.8712	0.9296***	0.7718***	-0.8911***	0.9044***	-0.1807***	-0.9655***	0.4469***	0.8222***	-0.4206***	1					
(16) <i>Govdown</i>	-0.0145	-0.0396	-0.0236	0.0179	-0.8093	0.5126***	0.8390***	-0.8162***	0.3575***	-0.4528***	-0.8191***	0.2305***	0.7774***	-0.6918***	0.6721***	1				
(17) <i>Lnta</i>	0.5820***	0.4929***	0.5564***	0.3010***	-0.0234	0.0243	0.0300	-0.0271	0.0223	-0.0303	0.0294	0.0376	-0.0299	0.0241	0.0271	0.0241	1			
(18) <i>ROA</i>	0.2775***	0.1542***	0.2574***	0.2046***	-0.0412	-0.0046	0.0328	-0.0351	-0.0116	-0.0609*	-0.0300	-0.0074	0.0477*	-0.0501***	0.0087	0.0565***	0.2906***	1		
(19) <i>DebtRatio</i>	0.0285	0.1388***	-0.0190	-0.0139	-0.0371	0.0329	0.0346	-0.0316	0.0310	-0.0125	-0.0373	0.0254	0.0426*	-0.0297	0.0325	0.0295	0.2879***	-0.1458***	1	
(20) <i>Mbratio</i>	0.0878***	0.0497***	0.0609***	0.0792***	-0.1314***	0.1350***	0.1282***	-0.1046***	0.1278***	-0.0415	-0.1441***	0.1097***	0.1613***	-0.0930***	0.1282***	0.1033***	-0.0943***	0.3292***	-0.1010***	1

Note: This table presents a Pearson's correlation analysis matrix of the variables. Panel A presents a correlation matrix for horizontal spillover variables. Panel B presents a correlation matrix for vertical spillover variables. *InESG*, *InEnv*, *InSoc*, and *InGov* denote the natural logarithms of the scores for ESG, Env-pillar, Soc-pillar, and Gov-pillar, respectively. *ESGpeer*, *Envpeer*, *Socpeer*, and *Govpeer* denote the peer average scores in the upstream, midstream, or downstream sector, excluding the focal company. *ESGrank*, *Envrank*, *Socrank*, and *Govrank* denote a company's score ranking in SASB-classified subsectors. *ESGgap*, *Envgap*, *Socgap*, and *Govgap* denote the natural logarithms of the peer average scores in the upstream sector. *ESGmid*, *Envmid*, *Socmid*, and *Govmid* denote the natural logarithms of the peer average scores in the midstream sector. *ESGdown*, *Envdown*, *Socdown*, and *Govdown* denote the natural logarithms of the peer average scores in the downstream sector. *Lnta* denotes the natural logarithm of total assets. *ROA* denotes the ratio of EBIT to average total assets. *DebtRatio* denotes the ratio of total debt to total assets. *Mbratio* denotes the ratio of the market value of equity to the book value of equity. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4
VIF values of variables in the regression model.

Variable	Horizontal spillovers		Vertical spillovers		
	VIF		Upstream VIF	Midstream VIF	Downstream VIF
<i>Peer_(t-1)</i>	1.01				
<i>Rank_(t-1)</i>	1.46				
<i>Up_(t-1)</i>				5.36	2.44
<i>Mid_(t-1)</i>			2.64		2.47
<i>Down_(t-1)</i>			2.61	5.35	
<i>Lnta_(t-1)</i>	1.76		1.37	1.31	1.31
<i>ROA_(t-1)</i>	1.36		1.31	1.36	1.37
<i>Dabratio_(t-1)</i>	1.01		1.16	1.16	1.16
<i>Mbratio_(t-1)</i>	1.22		1.21	1.20	1.21
Average VIF	1.33		1.72	2.62	1.66

Note: *Peer* denotes the natural logarithm of the peer average score in the upstream, midstream, or downstream sector, excluding the focal company. *Rank* denotes a company's score ranking in SASB-classified subsectors. *Up* denotes the natural logarithm of the peer average score in the upstream sector. *Mid* denotes the natural logarithm of the peer average score in the midstream sector. *Down* denotes the natural logarithm of the peer average score in the downstream sector. *Lnta* denotes the natural logarithm of total assets. *ROA* denotes the ratio of EBIT to average total assets. *DebtRatio* denotes the ratio of total debt to total assets. *Mbratio* denotes the ratio of the market value of equity to the book value of equity.

In general, the upstream and midstream sectors are highly interdependent and closely coordinated. In addition, the scale of midstream companies is larger than that of upstream companies, which results in a strong synchronization effect on the upstream sector. However, a major gap exists in product applications between the upstream and downstream sectors. Specifically, the applications of downstream consumer products are diverse and highly differentiated, complicating the integration of the upstream and downstream sectors. In addition, the average company size in the downstream sector is larger than that in the upstream sector, and the ESG needs in the downstream sector are more diverse. Consequently, synchronizing activities between companies in the downstream and upstream sectors is challenging. Consumers have increasingly shifted their attention toward CSR, and therefore, upstream companies may need to expend more effort to compensate for the shortcomings of downstream companies. The findings of the current and previous studies indicate that upstream suppliers face pressure from midstream customers that compels them to adopt more socially responsible practices (Dai et al., 2021; Schiller, 2018; She, 2022; Tao et al., 2022).

In the midstream sector, vertical spillovers affect ESG combined scores and Gov-pillar scores but not Env-pillar or Soc-pillar scores. These vertical spillovers originate from the downstream, and they have a negative effect on ESG combined scores but a positive effect on Gov-pillar scores. Midstream companies play a key role in Taiwan's semiconductor supply chain, and they generally have total assets and net revenues four to six times greater than those of upstream and downstream companies. Midstream companies also have substantial human and financial resources and robust corporate governance mechanisms. This may explain why their ESG combined

Table 5
Peer spillovers in supply chains.

Panel A. Horizontal spillovers	Upstream				Midstream				Downstream			
Variable	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov
<i>Peer</i> _(t-1)	1.5398 (1.00)	2.2861*** (2.68)	−0.4653 (−0.42)	−0.6807 (−0.41)	−0.9495 (−0.50)	0.5295 (0.36)	−2.0539 (−1.46)	−1.5506 (−0.72)	0.4330 (0.32)	−2.0848* (1.70)	0.3838 (0.38)	1.2403 (1.37)
<i>Rank</i> _(t-1)	0.1293*** (3.34)	−0.0284 (−0.35)	0.0485 (0.95)	0.1976*** (3.66)	0.1139* (1.81)	0.0456 (0.72)	0.1139 (0.92)	0.2947** (2.67)	0.0533** (2.12)	0.0215 (0.44)	0.1044* (2.51)	0.1263*** (3.72)
<i>Lnta</i> _(t-1)	0.0365** (2.02)	0.0080 (0.40)	0.0272* (1.66)	0.0643 (1.57)	0.0331 (0.94)	−0.0421 (−0.84)	0.0148 (0.58)	0.0534 (0.76)	0.0296*** (1.69)	−0.0154 (−0.61)	0.0123 (0.73)	0.0671* (1.69)
<i>ROA</i> _(t-1)	0.0006 (1.20)	0.0001 (−0.08)	−0.0007 (−1.34)	0.0028** (2.27)	−0.0011 (−1.06)	−0.0007 (−0.41)	0.0005 (0.70)	−0.0038 (−1.68)	0.0001 (0.09)	−0.0010 (−1.14)	0.0001 (−0.02)	0.0012 (0.92)
<i>Debratio</i> _(t-1)	−0.0007 (−1.14)	−0.0005 (−0.95)	−0.0007 (−1.03)	−0.0010 (−0.73)	−0.0018** (−3.32)	−0.0007 (−0.73)	0.0001 (0.18)	−0.0049** (−2.58)	−0.0002 (−0.49)	0.0008 (1.46)	0.0001 (0.06)	−0.0007 (−0.76)
<i>MBratio</i> _(t-1)	0.0038* (1.93)	0.0009 (0.89)	0.0029 (1.25)	0.0064 (1.24)	−0.0029 (−0.38)	0.0078 (0.76)	−0.0014 (−0.22)	−0.0082 (−0.56)	0.0018 (0.50)	0.0090* (1.82)	−0.0008 (−0.18)	0.0005 (0.07)
Constant	−2.5280 (−0.41)	−5.1858 (−1.56)	5.7020 (1.28)	6.0892 (0.93)	7.5557 (1.01)	2.2931 (0.40)	12.1789** (2.17)	9.7473 (1.11)	1.9720 (0.36)	12.3306** (2.51)	2.2684 (0.55)	−1.5908 (−0.45)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.6535	0.0406	0.6128	0.3635	0.6333	0.1150	0.7979	0.5005	0.5159	0.0045	0.7303	0.2827
F-statistic	2.53***	6.59***	5.57***	3.019***	2.47**	0.86	1.72	2.72**	1.18	2.09**	1.58	2.52***

scores surpass those of the other two industry stream sectors (Table 2). As leaders and benchmarks within the supply chain, midstream companies may face higher expectations in terms of their responsibility; if upstream and downstream companies do not adequately contribute in terms of CSR, midstream companies must compensate. This improves the ESG performance of the entire supply chain, and it may explain the negative vertical spillover effect observed for ESG combined scores.

Corporate governance is an internal control system that protects shareholder interests and is closely related to SSCM. Companies with effective corporate governance tend to consider the rights and interests of stakeholders other than shareholders, including employees, customers, and the general public (Luo & Tang, 2021; Tibiletti et al., 2021). Therefore, the positive spillover effect of downstream customers' Gov-pillar scores on midstream suppliers may arise from customers' concerns regarding potential value loss caused by their suppliers. In the current study, in the downstream sector, no significant vertical spillover effects were observed from the upstream and midstream sectors. However, a vertical spillover effect was observed in the upstream and midstream sectors for ESG combined scores, Soc-pillar scores, and Gov-pillar scores, with these effects diminishing in the downstream sector. In summary, our findings indicate that the forward industry stream (direct customers) in a supply chain influences the backward industry stream (suppliers); this finding is consistent with those in the literature regarding ESG spillovers between customers and suppliers (Dai et al., 2021; Huang et al., 2021; She, 2022; Tao et al., 2022).

The semiconductor industry consumes a considerable amount of energy. In consideration of the pressure companies in this industry are under to achieve zero emissions and comply with carbon taxes, identifying a means of enhancing the positive vertical spillover effects on Env-pillar scores is an urgent concern. In Taiwan, major midstream companies such as TSMC and UMC have assumed leadership roles in organizing supply chain alliances. These alliances encourage upstream and downstream peers to collaborate and engage in Env-pillar-related activities.

In this study, we observed a significantly positive correlation between the company size (*Lnta*) of upstream and downstream companies and their ESG combined scores, Soc-pillar scores, and Gov-pillar scores, although this correlation did not extend to midstream companies. Downstream companies with promising growth prospects (*MBratio*) tended to have high Env-pillar scores, whereas highly leveraged (*Debratio*) midstream companies tended to have low Soc-pillar and Gov-pillar scores. However, companies with high profitability (ROA) did not necessarily have high Env-pillar scores.

Panel B. Vertical spillovers	Upstream				Midstream				Downstream			
	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov
Variable												
$Up_{(t-1)}$					-2.0963 (-1.41)	-0.1027 (-0.15)	-1.2632 (-0.57)	21.4293 (1.57)	-0.2163 (-0.51)	0.3824 (0.96)	-0.2124 (-0.83)	-1.3413 (-0.51)
$Mid_{(t-1)}$	1.8012 (1.57)	-0.0399 (-0.18)	0.6777*** (3.13)	-0.3952 (-0.96)					-0.0610 (-0.27)	-0.0451 (-0.23)	-0.0983 (-0.40)	0.0185 (0.06)
$Down_{(t-1)}$	-9.3032 (-1.60)	0.6256 (1.13)	-2.1991*** (-2.86)	0.5820 (0.29)	-5.6340* (-1.70)	-1.2938 (-0.58)	-3.3416 (-0.43)	19.1954* (1.82)				
$Lnta_{(t-1)}$	0.0395* (1.75)	0.0078 (0.39)	0.0289 (1.61)	0.0648 (1.33)	0.0391 (1.09)	-0.0391 (-0.79)	0.0183 (0.62)	0.0830 (1.07)	0.0321* (1.81)	-0.0176 (-0.67)	0.0163 (0.90)	0.0717* (1.70)
$ROA_{(t-1)}$	0.0006 (1.04)	-0.0001 (-0.06)	-0.0007 (-1.33)	0.0028** (2.03)	-0.0009 (-0.71)	-0.0006 (-0.36)	0.0006 (0.63)	-0.0030 (-1.07)	0.0001 (0.21)	-0.0009 (-1.04)	0.0001 (0.13)	0.0011 (0.81)
$Debratio_{(t-1)}$	-0.0008 (-1.16)	-0.0005 (-0.94)	-0.0006 (-0.92)	-0.0013 (-0.86)	-0.0019* (-1.87)	-0.0006 (-0.63)	0.0001 (0.13)	-0.0052** (-2.15)	-0.0003 (-0.64)	0.0009 (1.59)	-0.0001 (-0.21)	-0.0010 (-0.97)
$MBratio_{(t-1)}$	0.0039* (1.84)	0.0011 (1.03)	0.0027 (1.18)	0.0074 (1.34)	-0.0004 (-0.05)	0.0084 (0.84)	-0.0008 (-0.12)	-0.0045 (-0.29)	0.0014 (0.41)	0.0099* (1.91)	-0.0012 (-0.28)	-0.0024 (-0.36)
Constant	33.5323* (1.81)	1.5688 (1.07)	9.8419*** (3.85)	2.7254 (0.37)	34.6380* (1.83)	9.9596** (1.95)	22.3439 (0.56)	-158.7597 (-1.66)	4.8343* (1.90)	2.7241 (1.19)	5.1129*** (2.7)	8.6917 (0.75)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.4095	0.1485	0.4690	0.1345	0.4102	0.1902	0.4812	0.1538	0.3160	0.0643	0.3012	0.0895
F-statistic	1.55	7.20***	7.35***	7.15***	1.89*	0.74	0.97	1.71	0.89	1.94*	0.54	1.44

Note: Panel A presents the results for horizontal spillovers. Panel B presents the results for vertical spillovers. ESG, Env, Soc, and Gov denote the natural logarithms of the scores of ESG, Env-pillar, Soc-pillar, and Gov-pillar, respectively. Peer denotes the natural logarithm of the peer average score in the upstream, midstream, or downstream sector, excluding the focal company. Rank denotes a company's score ranking in SASB-classified sub-sectors. Up denotes the natural logarithm of the peer average score in the upstream sector. Mid denotes the natural logarithm of the peer average score in the midstream sector. Down denotes the natural logarithm of the peer average score in the downstream sector. Lnta denotes the natural logarithm of total assets. ROA denotes the ratio of EBIT to average total assets. Debratio denotes the ratio of total debt to total assets. Mbratio denotes the ratio of the market value of equity to the book value of equity. Here, t-statistics (in parentheses) are calculated in accordance with White's (1980) heteroscedasticity-consistent standard errors. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

4.3. Moderating effect of network linkages on peer spillovers

When the number of network linkages increases, the competitive pressure and cooperation incentives from horizontal and vertical peers also increase. This indicates that network linkages may enhance peer spillover effects. Table 6 presents the descriptive statistics for subsamples classified by net revenue and operating cost ratio. Panels A and B present the descriptive statistics for the net revenue- and operating cost ratio-classified subsamples, respectively, and Panel C presents the scores of the subsamples sorted on the basis of

net revenue and operating cost ratio values. Each sample is divided into three groups: high, medium, and low. A value of 1 is the highest score. Analysis of the net revenue subsamples revealed that the *ESG* combined scores and three pillar scores increased as net revenue increased; in other words, companies with higher net revenues had higher *ESG* scores. By contrast, analysis of the operating cost ratio subsamples revealed that *Env*-pillar scores increased as the operating cost ratio increased, indicating that companies with higher operating cost ratios are more concerned with the *Env*-pillar. In addition, the scores of *ESG*, *Soc*-pillar, and *Gov*-pillar decreased with as operating cost ratios increased.

We examined the moderating effect of company network linkages on horizontal peer spillovers and *ESG* scores. Table 7 presents the regression results. Analysis of horizontal spillovers (Table 7, Panel A) revealed that network linkages had no moderating effect on horizontal peer spillovers for *ESG* combined scores, *Env*-pillar scores, and *Gov*-pillar scores. By contrast, high-revenue linkages had a moderating effect on *Soc*-pillar scores in the upstream and midstream sectors. Similar to the ranking spillover effect observed in the upstream sector, the peer spillover effect observed in the midstream sector was enhanced by high-revenue linkages. Analysis of vertical spillovers (Table 7, Panel B) for upstream companies in terms of *ESG* combined scores and *Env*-pillar scores revealed that high-revenue linkages had a positive moderating effect on midstream vertical spillovers but a negative moderating effect on downstream vertical spillovers. In terms of *Env*-pillar scores, high-cost linkages had a negative moderating effect on midstream spillovers but a positive moderating effect on downstream spillovers. In summary, because they are under pressure to achieve zero carbon emissions and comply with carbon taxes, upstream companies with high revenue and cost linkages have *Env*-pillar scores subject to a strong positive vertical peer effect from midstream (direct customers) and downstream (end customers) companies. For midstream and downstream companies, in terms of their *ESG* combined scores and *Env*-pillar scores, high-revenue linkages have a significant moderating effect on vertical peer spillovers, but high-cost linkages do not. High-revenue linkages reduce upstream vertical spillovers (suppliers) but promote downstream vertical spillovers (customers). In addition, network linkages do not have a moderating effect on the relationship between peer spillovers and *Gov*-pillar scores. Overall, the moderating effect of high-revenue linkages on the correlation between vertical peer spillovers and *ESG* scores is significant in the upstream and midstream sectors but not in the downstream sector.

Taiwan's semiconductor supply chain is unique in that it is dominated by midstream companies, such as TSMC, whose performance substantially affects the entire supply chain. In this study, we discovered that high-revenue linkages between the midstream sector and the other two industrial stream sectors enhance vertical spillovers. That is, network linkages enhanced the vertical spillover effect of the midstream sector. Consistent with previous research on customer relationship spillovers (Dai et al., 2021; Lian et al., 2023; Tang et al., 2023), we discovered that high-revenue linkages enhance vertical spillovers. Although we did not observe a moderating effect of high-cost linkages on peer spillovers, considering the pressures companies face to achieve zero carbon emissions and comply with carbon taxes, we expect cost linkages to become a key moderating factor in the future.

4.4. Additional test

Multiple studies have reported a cause-and-effect relationship between a company's score and the scores of its competitors and

Table 6
Descriptive statistics for net revenue and operating cost ratio subsamples.

Panel A. Net revenue		Low				Medium				High			
Variable		Mean	Median	Min	Max	Mean	Median	Min	Max	Mean	Median	Min	Max
<i>ESG</i>		50.47	49.75	35.44	70.44	54.54	57.28	38.30	70.35	59.29	58.28	42.51	80.34
<i>Env</i>		48.54	45.39	41.42	72.42	53.17	52.68	32.53	79.71	59.45	59.80	32.60	88.61
<i>Soc</i>		51.71	52.33	36.15	72.54	54.58	54.78	37.91	73.83	61.51	62.29	39.91	91
<i>Gov</i>		50.46	50.04	22.45	78.18	55.37	56.85	23.71	78.23	57.05	56.91	25.23	77.92
Net revenue (NT\$ million)		595	538	74	1323	273	2600	1173	5389	53900	13800	4653	533000
Operating cost ratio		65.84	66.86	0.32	109.06	74.42	75.67	0.32	109.06	77.63	78.84	39.62	104.06
Panel B. Operating cost ratio		Low				Medium				High			
Variable		Mean	Median	Min	Max	Mean	Median	Min	Max	Mean	Median	Min	Max
<i>ESG</i>		55.74	54.69	35.44	80.01	54.89	54.08	37.88	80.34	53.68	52.83	35.95	78.41
<i>Env</i>		53.45	48.92	33.85	85.11	53.79	52.94	32.53	88.61	53.93	52.99	32.6	85.95
<i>Soc</i>		57.14	56.13	36.15	91	56.20	55.53	37.41	89.13	54.46	53.23	36.89	84.1
<i>Gov</i>		55.73	55.23	22.45	78.23	54.36	55.34	23.71	77.36	52.81	53.96	23.06	77.92
Net revenue (NT\$ million)		13400	1255	74	533000	11200	2807	74	283000	32500	3884	74	533000
Operating cost ratio		52.99	57.26	0.32	68.02	74.04	73.85	65.33	83.05	90.84	90.00	79.39	109.06
Panel C. Score sorting of subsamples		Net revenue				Operating cost ratio							
Group		<i>ESG</i>	<i>Env</i>	<i>Soc</i>	<i>Gov</i>	<i>ESG</i>	<i>Env</i>	<i>Soc</i>	<i>Gov</i>				
Low		3	3	3	3	1	3	1	1				
Medium		2	2	2	2	2	2	2	2				
High		1	1	1	1	3	1	3	3				

Note: Panel A presents the descriptive statistics for a net revenue subsample. Panel B presents the descriptive statistics for a operating cost ratio subsample. Panel C presents the score sorting of the net revenue and operating cost ratio subsamples. On the basis of the net revenue and operating cost ratio values, we divided each sample into three groups: high, medium, and low. A value of 1 represents the highest score. *ESG*, *Env*, *Soc*, and *Gov* denote the *ESG* combined score, *Env*-pillar score, *Soc*-pillar score, and *Gov*-pillar score, respectively. Operating cost ratio denotes the ratio of operating cost to net revenue.

Table 7

Moderating effect of network linkages on peer spillovers.

Panel A. Horizontal spillovers	Upstream				Midstream				Downstream			
Variable	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov
$Peer_{(t-1)}$	1.5600 (0.98)	2.2840*** (2.64)	−0.2910 (−0.26)	−0.5886 (−0.36)	−1.3913 (−0.74)	0.7866 (0.51)	−1.7959 (−1.31)	−1.9494 (−0.93)	0.4522 (0.34)	−2.1661* (−1.85)	0.3303 (0.32)	1.2990 (1.44)
$Rank_{(t-1)}$	0.1341*** (2.93)	0.0203 (0.28)	0.0218 (0.54)	0.2342*** (3.59)	0.1114* (1.69)	0.1669 (2.01)	0.2738*** (3.83)	0.3297*** (2.78)	0.0653** (1.98)	0.0894 (1.49)	0.1059** (2.57)	0.1823*** (3.62)
$Peer_{(t-1)} \times$ $highsale_{(t-1)}$	−0.0023 (−0.09)	0.0080 (0.66)	−0.0639** (−2.58)	0.0332 (1.07)	0.0137 (0.81)	0.0136 (1.07)	0.0521** (2.20)	0.0408 (1.06)	0.0027 (0.57)	0.0105 (1.51)	0.0072 (1.33)	0.0042 (0.44)
$Rank_{(t-1)} \times$ $highsale_{(t-1)}$	0.0110 (0.09)	−0.0570 (−0.79)	0.3386*** (2.84)	−0.1274 (−1.17)	−0.0490 (−0.48)	−0.1158 (−1.55)	−0.2826** (−2.21)	−0.1639 (−0.73)	−0.0273 (−0.86)	−0.0996 (−1.60)	−0.0533 (−1.27)	−0.0337 (−0.65)
$Peer_{(t-1)} \times$ $highcost_{(t-1)}$	0.0051 (0.56)	0.0011 (0.35)	0.0103 (0.84)	0.0092 (0.70)	−0.0055 (−0.60)	0.0391 (3.19)	0.0127 (1.06)	−0.0152 (−0.57)	−0.0005 (−0.07)	−0.0037 (−0.54)	−0.0088 (−1.63)	0.0109 (0.73)
$Rank_{(t-1)} \times$ $highcost_{(t-1)}$	−0.0335 (−0.88)	−0.0012 (−0.07)	−0.0733 (−1.05)	−0.0751 (−0.94)	0.0587 (1.26)	−0.1418 (−1.54)	−0.0483 (−0.77)	0.0541 (0.41)	−0.0131 (−0.31)	0.0037 (0.07)	0.0424 (1.51)	−0.0914 (−1.33)
Constant	−2.6100 (−0.41)	−5.1893 (−1.54)	4.9059 (1.07)	5.7594 (0.88)	9.3169 (1.25)	1.1604 (0.19)	11.0684* (2.00)	11.3143 (1.32)	1.9015 (0.35)	12.6288*** (2.70)	2.5064 (0.60)	−1.8371 (−0.52)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.6586	0.0575	0.5594	0.3354	0.6437	0.0779	0.6583	0.4863	0.5267	0.0035	0.6760	0.3026
F-statistic	2.27***	4.82***	4.40***	2.91***	2.65***	3.22***	2.35**	3.26***	1.08	2.92***	1.83**	2.48***

Panel B. Vertical spillovers	Upstream				Midstream				Downstream			
Variable	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov
$Up_{(t-1)}$					−0.886 (−0.71)	0.7010 (0.63)	−0.8044 (−0.34)	23.5441* (1.79)	−0.1151 (−0.26)	0.4903 (1.21)	−0.0648 (−0.22)	−1.2831 (−0.49)
$Mid_{(t-1)}$	1.4923 (1.30)	−0.3789*** (−2.74)	0.8672*** (3.85)	−0.6260 (−1.20)					−0.1656 (−0.69)	−0.1632 (−0.65)	−0.2566 (−0.90)	0.0102 (0.03)
$Down_{(t-1)}$	−8.7903 (−1.53)	1.0067** (2.19)	−2.3291*** (−3.02)	0.9766 (0.49)	−6.5845* (−1.77)	−1.9076* (−1.84)	−4.5205 (−0.59)	17.2441* (1.70)				
$Up_{(t-1)} \times$ $highsale_{(t-1)}$					−1.4714* (−1.70)	−2.0543 (−1.56)	−1.4244*** (−2.87)	−1.1980 (−0.70)	−0.1823 (−1.02)	−0.4617* (−1.87)	−0.2642 (−1.09)	0.3461 (0.83)
$Mid_{(t-1)} \times$ $highsale_{(t-1)}$	1.2365** (2.10)	2.2445*** (3.15)	−1.0309*** (−2.63)	1.0913 (1.32)					0.1799 (1.02)	0.4495* (1.87)	0.2628 (1.10)	−0.3463 (−0.84)
$Down_{(t-1)} \times$ $highsale_{(t-1)}$	−1.2510** (−2.09)	−2.3011*** (−3.15)	1.0424*** (2.62)	−1.0741 (−1.30)	1.4808* (1.70)	2.0505 (1.57)	1.4412*** (2.87)	1.2174 (0.71)				
$Up_{(t-1)} \times$ $highcost_{(t-1)}$					−1.0124 (−0.97)	0.5224 (0.48)	0.0802 (0.17)	−3.3442 (−1.66)	−0.0658 (−0.39)	0.1214 (0.54)	−0.0994 (−0.42)	−0.2913 (−0.73)
$Mid_{(t-1)} \times$ $highcost_{(t-1)}$	−0.0326 (−0.04)	−0.9395*** (−3.00)	0.6512 (0.91)	−0.4448 (−0.42)					0.0626 (0.38)	−0.1214 (−0.55)	0.0958 (0.41)	0.2884 (0.72)

(continued on next page)

Table 7 (continued)

Panel B. Vertical spillovers	Upstream				Midstream				Downstream			
	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov
$Down_{(t-1)} \times$ $highcost_{(t-1)}$	0.0328 (0.04)	0.9631*** (3.01)	−0.6612 (−0.90)	0.4449 (0.42)	1.0202 (0.97)	−0.5033 (−0.46)	−0.0728 (−0.16)	3.3369 (1.66)				
Constant	32.7656* (1.79)	1.5016 (1.02)	9.5689*** (3.65)	2.1775 (0.30)	33.4439* (1.74)	9.0682* (1.87)	25.0976 (0.63)	−159.5410* (−1.74)	4.8478 (1.92)	2.8016 (1.21)	5.1630*** (2.70)	8.4502 (0.73)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.4067	0.0099	0.4071	0.1094	0.4570	0.0955	0.4876	0.1707	0.3268	0.0691	0.3037	0.0912
F-statistic	2.62**	6.21***	5.93***	1.54	1.44	0.86	2.46**	1.71*	0.97	2.65***	0.79	1.68*

Note: Panel A presents the results for horizontal spillovers. Panel B presents the results for vertical spillovers. *ESG*, *Env*, *Soc*, and *Gov* denote the natural logarithms of the *ESG* combined score, *Env*-pillar score, *Soc*-pillar score, and *Gov*-pillar score, respectively. *Peer* denotes the natural logarithm of the peer average score in the upstream, midstream, or downstream sector, excluding the focal company. *Rank* denotes a company's score ranking in SASB-classified subsectors. *Up* denotes the natural logarithm of the peer average score in the upstream sector. *Mid* denotes the natural logarithm of the peer average score in the midstream sector. *Down* denotes the natural logarithm of the peer average score in the downstream sector. The value of *highsale* (*highcost*) is 1 if company's net revenue (operating cost ratio) is the top 1/3 of all companies and 0 otherwise. Controls include *Lnta*, *ROA*, *Debratio*, and *MBratio*. *Lnta* denotes the natural logarithm of total assets. *ROA* denotes the ratio of EBIT to average total assets. *Debratio* denotes the ratio of total debt to total assets. *MBratio* denotes the ratio of the market value of equity to the book value of equity. Here, *t*-statistics (in parentheses) are calculated in accordance with White's (1980) heteroscedasticity-consistent standard errors. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 8

Interactive effects of positive or negative score changes on peer spillovers.

Panel A. Horizontal spillovers	Upstream				Midstream				Downstream			
Variable	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov
$Positive_{(t-1)}$	3.7561 (0.44)	20.7792** (1.89)	−7.3397 (−0.73)	2.4982 (0.71)	11.7361 (1.42)	31.8621*** (2.71)	20.0923* (1.97)	25.1430* (1.70)	−2.6976 (−0.41)	9.6674 (1.60)	8.6436* (1.89)	−2.4018 (−0.40)
$Negative_{(t-1)}$	−16.4940** (−2.25)	−9.7451*** (−2.75)	3.9816 (0.76)	(omitted)	−11.3660 (−1.29)	(omitted)	−0.7280 (0.11)	(omitted)	−1.1522 (−0.20)	3.0692 (0.74)	(omitted)	(omitted)
$Peer_{(t-1)} \times Positive_{(t-1)}$	−0.9451 (−0.44)	−5.1984* (−1.89)	1.8031 (0.73)	−1.1982 (−0.51)	−2.9207 (−1.42)	−4.6591* (−1.74)	−5.0055* (−1.99)	−4.7152** (−2.31)	0.6595 (0.41)	−2.4278 (−1.60)	−0.7484 (−0.62)	1.5727 (1.27)
$Rank_{(t-1)} \times Positive_{(t-1)}$	0.1188** (2.57)	−0.1563* (−1.67)	0.0517 (0.76)	0.1037 (1.51)	0.1338 (1.49)	−0.0221 (−0.29)	0.1453 (1.28)	0.1758 (1.21)	0.0409 (1.32)	−0.0280 (−0.57)	0.0718 (1.46)	0.1269*** (2.88)
$Peer_{(t-1)} \times Negative_{(t-1)}$	4.1021** (2.25)	2.4572*** (2.77)	−0.9904 (−0.76)	−0.5868 (−0.25)	2.8438 (1.30)	3.3348* (1.73)	0.1412 (0.09)	1.5594 (0.48)	0.2731 (0.19)	−0.7816 (−0.75)	1.3826 (1.33)	0.9716 (0.87)
$Rank_{(t-1)} \times Negative_{(t-1)}$	0.0917** (2.05)	−0.0883 (−1.12)	0.0429 (0.81)	0.1148*** (1.76)	0.0918 (1.46)	0.0001 (0.00)	0.0777 (0.66)	0.2611** (2.1)	0.0314 (1.21)	0.0247 (0.42)	0.0656 (1.39)	0.1173*** (3.16)
Constant	3.7030*** (26.88)	3.9456*** (33.74)	3.8379*** (33.60)	5.7175 (0.60)	3.6564*** (11.49)	−8.9991 (−0.57)	3.8836*** (21.53)	−2.7905 (−0.21)	3.7809*** (27.57)	4.0722*** (24.36)	−1.7529 (−0.42)	−0.5000 (−0.11)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.6260	0.3816	0.5978	0.2900	0.6272	0.0692	0.7095	0.4485	0.4800	0.0095	0.6603	0.2834
F-statistic	13.21***	11.51***	4.22***	4.74***	3.16***	2.90***	28597.73***	4.35***	2667.88***	3.30***	2.12**	1.92**
Panel B. Vertical spillovers	Upstream				Midstream				Downstream			
Variable	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov
$Positive_{(t-1)}$	10.7279 (0.57)	26.9931*** (6.62)	−0.5859 (−0.11)	−12.4667* (−1.83)	42.7254* (1.72)	11.8759 (1.37)	−11.3478 (−0.28)	−5.2787 (−0.27)	5.4757* (1.76)	−3.7218 (−0.42)	4.5339* (1.87)	−2.7523 (−0.55)
$Negative_{(t-1)}$	33.5146** (1.94)	30.2131*** (11.69)	13.3316*** (3.30)	(omitted)	39.0899* (1.78)	(omitted)	2.9142 (0.06)	(omitted)	−2.2674 (−0.68)	−23.5956*** (−6.08)	(omitted)	(omitted)
$Up_{(t-1)} \times Positive_{(t-1)}$					−3.3373* (−1.76)	−1.3918 (−1.27)	−0.2173 (−0.09)	36.0668** (2.03)	−1.1024** (−2.21)	0.0058 (0.00)	−1.0694*** (−3.27)	−1.2924 (−0.47)
$Mid_{(t-1)} \times Positive_{(t-1)}$	1.7292 (1.39)	−0.2014 (−0.41)	−0.1678 (−0.21)	−0.2018 (0.32)					−0.2652 (−0.83)	0.9120 (0.90)	0.0665 (0.17)	−0.0505 (−0.10)
$Down_{(t-1)} \times Positive_{(t-1)}$	−4.4483 (−0.75)	−6.6090*** (−6.78)	0.3219 (0.32)	2.5745 (1.10)	−7.3426 (−1.65)	−1.8663 (−1.06)	3.0243 (0.39)	33.0863** (2.30)				
$Up_{(t-1)} \times Negative_{(t-1)}$					−1.4018 (−0.93)	1.1067 (1.60)	0.8220 (0.30)	37.8570** (2.14)	0.4724 (0.87)	3.4078*** (6.90)	0.3928 (1.49)	−2.0472 (−0.76)
$Mid_{(t-1)} \times Negative_{(t-1)}$	1.5431 (1.38)	−1.7536*** (−6.27)	−0.3295 (−0.47)	−0.4559 (−0.76)					0.0871 (0.27)	2.4775*** (4.93)	−0.2754 (−0.85)	0.0106 (0.03)

(continued on next page)

Table 8 (continued)

Panel B. Vertical spillovers	Upstream				Midstream				Downstream			
Variable	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov	ESG	Env	Soc	Gov
$Down_{(t-1)} \times$	−9.9788*	−5.8396***	−3.0006***	−0.3096	−8.3752**	−1.3811	−1.6018	29.9582**				
$Negative_{(t-1)}$	(1.84)	(−6.96)	(−3.67)	(−0.15)	(−2.05)	(−1.19)	(−0.17)	(2.20)				
Constant	3.7541***	4.0892***	3.8000***	6.4956	3.6338***	5.4089	4.0340	−267.2543	3.7924***	4.1500***	3.4434	11.5601
	(25.48)	(29.95)	(30.46)	(1.48)	(12.65)	(0.88)	(22.58)	(−2.16)	(28.83)	(21.71)	(1.62)	(0.98)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.4316	0.0997	0.4789	0.1595	0.4511	0.0730	0.4274	0.1756	0.3403	0.0316	0.2506	0.0990
F-statistic	2.35***	2.05***	2.55***	4.40***	2.65***	2.26**	2.23***	2.49***	203856.82***	1.22***	3.30***	1.24

Note: Panel A presents the results for horizontal spillovers. Panel B presents the results for vertical spillovers. *ESG*, *Env*, *Soc*, and *Gov* denote the natural logarithms of the *ESG* combined score, *Env*-pillar score, *Soc*-pillar score, and *Gov*-pillar score, respectively. *Peer* denotes the natural logarithm of the peer average score in the upstream, midstream, or downstream sector, excluding the focal company. *Rank* denotes a company's score ranking in SASB-classified subsectors. *Up* denotes the natural logarithm of the peer average score in the upstream sector. *Mid* denotes the natural logarithm of the peer average score in the midstream sector. *Down* denotes the natural logarithm of the peer average score in the downstream sector. *Positive* = 1 if score change >0 and 0 otherwise. *Negative* = 1 if score change <0 and 0 otherwise. Controls include *Lnta*, *ROA*, *Debratio*, and *MBratio*. *Lnta* denotes the natural logarithm of total assets. *ROA* denotes the ratio of EBIT to average total assets. *Debratio* denotes the ratio of total debt to total assets. *MBratio* denotes the ratio of the market value of equity to the book value of equity. Here, *t*-statistics (in parentheses) are calculated in accordance with White's (1980) heteroscedasticity-consistent standard errors. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively. Some values are omitted because of collinearity.

Table 9
Industry concentration in Taiwan's semiconductor supply chain.

	Upstream		Midstream		Downstream	
Concentration	Total Assets	Net Revenue	Total Assets	Net Revenue	Total Assets	Net Revenue
CR3	0.5894	0.4703	0.8034	0.7791	0.4687	0.6066
HHI	0.1745	0.1392	0.4015	0.3538	0.0996	0.1690
No. of companies	101		44		126	

Note: This table represents the industry concentration of Taiwan's semiconductor supply chain from 2015 to 2020. CR3 is the average concentration ratio of the top three companies from 2015 to 2020. HHI is the average value from 2015 to 2020.

customers (Cao et al., 2019; Dai et al., 2021; Lian et al., 2023; Tang et al., 2023). In this study, we examined the interactive effect of a company's score change on peer spillovers. We calculated the annual change observed in the ESG combined score and three pillar scores and used two dummy variables to determine the direction of score change (i.e., *Positive* or *Negative*). *Positive* was assigned a value of 1 if the score change was greater than 0 and a value of 0 otherwise, and *Negative* was assigned a value of 1 if the score change was smaller than 0 and a value of 0 otherwise. The models are as follows:

$$Score_{it} = \alpha_0 + \beta_1 Positive_{it-1} + \beta_2 Negative_{it-1} + \sum \theta_n Peer_{jt-1} \times (Positive|Negative)_{it-1} + \sum \vartheta_n Rank_{qt-1} \times (Positive|Negative)_{it-1} + \sum \delta_k Control_{kt-1} + \theta_t + \mu_i + \varepsilon_{it} \quad (5)$$

$$Score_{it} = \alpha_0 + \beta_1 Positive_{it-1} + \beta_2 Negative_{it-1} + \sum \gamma_j (Up|Mid|Down)_{jt-1} \times (Positive|Negative)_{it-1} + \sum \delta_k Control_{kt-1} + \theta_t + \mu_i + \varepsilon_{it} \quad (6)$$

where $Positive_{it-1}$ and $Negative_{it-1}$ represent the score changes of a company i in year $t-1$. The other variables are the same as those in Eqs. (1) and (2).

Table 8 presents the regression results. As indicated in Panel A, positive peer spillovers were enhanced for upstream and midstream companies with negative *Env*-pillar score changes but not for companies with positive score changes. Companies with decreasing scores experienced increasing peer competition pressure and increasing spillover effects from horizontal peers. Therefore, these companies tended to focus on improving their *Env*-pillar performance. Downstream companies had highly diversified and differentiated products. Therefore, the effect of score changes on horizontal spillovers was not significant in these companies. For midstream companies with high *Soc*-pillar and *Gov*-pillar scores, the effects of horizontal peer spillovers decreased. In addition, for downstream companies with high or low *Gov*-pillar scores, horizontal rank spillovers had a major influence because of the effect of score ranking.

As indicated in Panel B of Table 8, upstream companies with low *Env*-pillar scores experienced a weak vertical spillover effect, whereas downstream companies with low *Env*-pillar scores experienced a strong vertical spillover effect. In Taiwan's semiconductor supply chain, most upstream companies are small. Companies with low *Env*-pillar scores may struggle with limited funding and resources. Because they do not have a direct connection to end consumers, these companies face minimal pressure and do not have strong incentives to improve themselves. Therefore, in the upstream sector, low *Env*-pillar scores do not enhance the effects of vertical spillovers originating from the midstream and downstream sectors. Nevertheless, end consumers tend to pay considerable attention to environmental topics and select products and companies that support environmental protection. Overall, downstream companies with low *Env*-pillar scores typically experience adverse effects on their future profit performance. Therefore, by ensuring a green supply chain, companies can foster a positive brand image and customer loyalty and improve their market competitiveness and value. This can enhance the vertical spillover effects of *Env*-pillar in the upstream and midstream sectors. In addition, midstream companies with positive or negative score changes in *Gov*-pillar typically experience an increased vertical peer spillover effect, indicating that score changes in *Gov*-pillar in the midstream sector closely relate to vertical peers' *Gov*-pillar in supply chains. The multiple major midstream companies in Taiwan, such as TSMC and UMC, have actively encouraged their suppliers and customers (upstream and downstream sectors) to form sustainable supply chain alliances and engage in zero carbon emission efforts.

Table 9 presents the concentration ratios of the top three companies (CR3) and their Herfindahl–Hirschman Index (HHI)²⁰ values expressed in terms of the total assets and net revenues of Taiwan's semiconductor supply chain from 2015 to 2020. According to the

²⁰ Top n largest companies' concentration ratio and HHI are indicators used to measure market concentration. The top n largest companies' concentration ratio is defined as the market share of the top n largest companies in the market. The HHI is calculated by squaring the market share of each company competing in the market and then summing the results. Here, $CR_n = \sum_{i=1}^n s_i$, and $HHI = \sum_{i=1}^n s_i^2$, where s_i is the market share of a company i .

Table 10

Moderating effect of industry concentration on peer spillovers in the midstream sector.

Panel A. Horizontal spillovers	ESG		Env		Soc		Gov	
Variable	HHI_A	HHI_R	HHI_A	HHI_R	HHI_A	HHI_R	HHI_A	HHI_R
$Peer_{(t-1)} \times HHI_{(t-1)}$	-177.9264 (-1.25)	-147.6559 (-1.29)	-240.6368** (-2.10)	-155.0713 (-1.24)	-336.9430** (-2.03)	-78.5594 (-0.67)	217.8283 (1.55)	278.2926* (2.01)
Panel B. Vertical spillovers	ESG		Env		Soc		Gov	
Variable	HHI_A	HHI_R	HHI_A	HHI_R	HHI_A	HHI_R	HHI_A	HHI_R
$Up_{(t-1)} \times HHI_{(t-1)}$	108.5937* (1.80)	157.2348** (2.46)	80.6325* (1.85)	0.0605 (0)	-39.15589 (-1.28)	-36.1505 (-1.06)	635.8776 (1.63)	122.2658* (1.78)
$Down_{(t-1)} \times HHI_{(t-1)}$	-109.0030* (-1.80)	-158.1516** (2.46)	-80.0783* (-1.84)	-0.0268 (0)	39.82963 (1.28)	36.2617 (1.05)	-627.7639 (1.63)	-122.2756* (-1.78)

Note: Panel A presents the results of horizontal spillovers. Panel B presents the results of vertical spillovers. Only partial results for HHI interaction terms are depicted, with the other variables omitted. HHI_A represents the HHI measured by total assets. HHI_R represents the HHI measured by net revenue. $Peer$ denotes the natural logarithm of the peer average score in the upstream, midstream, or downstream sector, excluding the focal company. Up denotes the natural logarithm of the peer average score in the upstream sector. $Down$ denotes the natural logarithm of the peer average score in the downstream sector. Here, t -statistics (in parentheses) are calculated in accordance with White's (1980) heteroscedasticity-consistent standard errors. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

HHI values of total assets and net revenues, the upstream and downstream sectors had a low level of industry concentration (i.e., diversified industry segmentation).²¹ By contrast, the midstream sector had a high level of industry concentration. A total of 44 midstream companies were identified in the sample, with the average concentration ratios of the total assets and net revenues of the top three companies from 2015 to 2020 being 80.34 % and 77.91 %, respectively.²² These results indicate a high level of industry concentration in the midstream sector.

According to the literature, companies tend to positively respond to their competitors' decision-making behaviors as a means of coping with competition, which in turn enhances the CSR peer effect (Cao et al., 2019; Liu & Wu, 2016). Market competition is a key driver of companies' ESG practices (Martins, 2022), and competitive imitation is a potential cause of the peer effect reported to affect ESG performance (Zhao & Wang, 2024). In this study, to determine whether industry concentration had a moderating effect on horizontal and vertical ESG spillovers in supply chains, we used the HHI as a proxy for the level of industry concentration²³ and added the interaction term of the HHI in Eqs. (1) and (2). Because Rank represents a company's score ranking in its SASB subsector and the HHI is typically calculated for the semiconductor industry, Eq. (1) does not include the variables of $Rank_{qt-1}$ and $Rank_{qt-1} \times HHI_{jt-1}$. The models are as follows:

$$Score_{it} = \alpha_0 + \beta_1 Peer_{jt-1} + \beta_2 Peer_{jt-1} \times HHI_{jt-1} + \sum \delta_k Control_{kt-1} + \theta_t + \mu_i + \varepsilon_{it} \quad (7)$$

$$Score_{it} = \alpha_0 + \sum \beta_j (Up|Mid|Down)_{jt-1} + \sum \gamma_j (Up|Mid|Down)_{jt-1} \times HHI_{jt-1} + \sum \delta_k Control_{kt-1} + \theta_t + \mu_i + \varepsilon_{it} \quad (8)$$

where HHI_{jt-1} represents the level of industry concentration of industry stream j in year $t-1$. Here, $HHI = \sum_{i=1}^n w_{it}^2$, where w_{it} is the weight of the total assets (net revenue) of each company by industry stream and year (referred to as HHI_A , HHI_R), and ranges between 0 and 1.

In Taiwan, the semiconductor supply chain focuses on wafer manufacturing in the midstream sector. This likely explains why our results indicate that the moderating effects of the HHI on the upstream and downstream sectors are not significant. Table 10 presents only results for the midstream sector, with the results for the upstream and downstream sectors omitted. As presented in Panel A, when the level of industrial concentration increases, the horizontal peer spillover effects of *Env*-pillar and *Soc*-pillar scores decrease, with a positive moderating effect on *Gov*-pillar scores. These results indicate that for large companies with major market shares, their *Env*-pillar and *Soc*-pillar scores serve as benchmarks for their peers (Zhao & Wang, 2024). In other words, their performance in the *Env*-pillar and *Soc*-pillar affects their peers. Therefore, when a company's industry concentration increases, it becomes less affected by

²¹ Taiwan's semiconductor supply chain comprises 22 subindustries, with 16 in the upstream sector, 6 industries in the midstream sector, and 6 industries in the downstream sector (see Appendix A1). Some companies in the semiconductor supply chain are also in other supply chains, belonging to 13 SASB subindustries (see Appendix A2). Given the heterogeneity of these subindustries, the HHI may not accurately represent market competition in Taiwan's semiconductor supply chain. In Taiwan's semiconductor industry, the HHI reflects only the level of industry concentration, and a high level of industry concentration does not necessarily indicate high industry competition. In this study, we used the term *industry concentration* instead of *industry competition*.

²² The top three companies in the midstream sector were TSMC, UMC, and Delta Electronics.

²³ CR_n is the sum of the weights of the top n largest companies in the market. Its value does not account for the effect of small companies and company size distribution. The HHI is the sum of the squares of the weights of each company; it therefore overcomes the shortcomings of CR_n and is sensitive to changes in market structure.

peers who are not leaders, indicating that leaders or core companies in the industry play a key role in ESG peer spillover effects.²⁴

As indicated by Panel B in Table 10, any increase in a company's industry concentration in the upstream sector can yield additional forward spillovers for ESG combined scores, *Env*-pillar scores, and *Gov*-pillar scores in the midstream sector. However, the industry concentration in the downstream sector has a negative effect on backward spillovers in the midstream sector.²⁵ Because the industry is shifting toward green supply chains, the concentration of upstream industries (e.g., high-market-share companies) is likely to become a key driver of ESG forward spillovers to peers (Martins, 2022). When the downstream concentration (low competition) increases, the effect of ESG backward spillovers on the midstream sector is likely to decrease. In other words, in a product oligopoly market, pressure from end consumers on companies to achieve high ESG performance decreases.

5. Conclusion

Many governments, companies, and investors have started to focus on various ESG factors. These factors influence all aspects of business operations, and companies with high ESG performance often enjoy additional market advantages and business opportunities. Most of the literature on this topic has focused on the influence of customer–supplier relationships on ESG peer spillovers, with studies on horizontal and vertical peer spillovers in supply chains being limited. In this study, we used data from Taiwan's semiconductor industry to determine how ESG peer spillovers propagate along supply chains through horizontal and vertical peers.

Our results indicate that the horizontal spillover effect of peers' average scores is more pronounced in the upstream sector and less pronounced in the midstream and downstream sectors. In addition, the forward industry stream (direct customers) has a positive vertical spillover effect on the backward industry stream (suppliers). Our findings also demonstrate that high-revenue linkages have a positive moderating effect on peer spillovers that increases in midstream vertical spillovers. Because of the moderating effect of network linkages, Taiwan's semiconductor supply chain has a vertical spillover effect on *Env*-pillar and *Soc*-pillar scores. Although we did not observe high-cost linkages to have a moderating effect on peer spillovers, given current trends toward achieving zero carbon emissions and complying with carbon taxes, *Env*-pillar is expected to be a challenge for companies, and therefore, cost linkages are likely to be a key moderator in the future. Companies with negative score changes typically experience an increase in horizontal peer spillovers and a decrease in vertical peer spillovers. Additionally, industry concentration has a moderating effect on the semiconductor supply chain in the midstream sector. When the concentration is higher, leaders and core companies tend to play a key role in ESG horizontal peer spillover effects. In the upstream sector, any increase in industry concentration increases the effect of ESG forward spillovers on the midstream sector, whereas in the downstream sector, any increase in industry concentration reduces the effect of ESG backward spillovers on the midstream sector.

In summary, horizontal and vertical ESG peer spillovers are prevalent in Taiwan's semiconductor supply chain. In this supply chain, core and leader companies, such as TSMC, play a key role. They also actively engage in ESG activities, establish ESG requirements for customers and suppliers, and have spillover effects on peers in the supply chain. Many electronic manufacturer supply chains, such as those of Apple, NVIDIA, and SMC have similar characteristics to those of semiconductor supply chains. Therefore, the ESG peer spillovers identified in our study are likely to also be prevalent in these global supply chains. In our study, we discovered that transaction linkages have a moderating effect on ESG peer spillovers. Given that the semiconductor industry comprises many sub-industries, ESG peer spillovers may spread to non-semiconductor industries through transaction network linkages between suppliers and customers. In recent years, TSMC has expanded its production facilities to the United States, Germany, Japan, and other countries. The ESG peer spillover effects in Taiwan's semiconductor supply chain may affect manufacturers in other countries through TSMC's global supply chain. Overall, our findings can be used as a reference for governments wishing to formulate policies that promote ESG peer spillover effects in supply chains. In consideration of the pressure of ESG vertical peer spillovers in supply chains, companies should actively engage in ESG activities to meet the requirements of SSCM.

Author statement

All authors have seen and approved the final version of the manuscript being submitted. We warrant that the article is our original work, hasn't received prior publication and isn't under consideration for publication elsewhere. In addition, there's no financial/personal interest or belief that could affect our objectivity.

Declarations of interest

None.

²⁴ TSMC and UMC actively encourage their suppliers and customers to form sustainable supply chain alliances and to engage in zero carbon emission efforts. For example, TSMC assists its suppliers in establishing environmental protection systems for emission reduction and encourages them to install carbon-capturing equipment. UMC has established the Triple R Alliance to lead suppliers in recycling reduction and energy saving.

²⁵ We replaced the HHI with CR_3 and obtained the same empirical results for horizontal and vertical spillovers.

Appendices

Appendix A1

Categories covered in Taiwan's semiconductor supply chain.

Upstream (SIP/IC design)	Midstream (IC/wafer fabrication)	Downstream (IC testing and packaging/module)
IP design/IC design OEM service	Other IC/diode manufacturing	Production process and testing equipment
Disk storage controller IC	IC/wafer manufacturing	IC packaging test
Flat panel display driver IC	DRAM manufacturing	IC module
I/O interface IC	Wafer manufacturing	IC channel
Optical storage control IC	Chemicals	Lead frame
LED driver IC	Mask	Substrate
Memory control IC		
Internet communication IC		
Image sensing IC		
Microcontroller IC		
Power management IC		
Light source management IC		
Memory IC		
Consumer IC		
Optical communication IC		
IC design		

Note: Industry Value Chain Information Platform (<https://ic.tpex.org.tw/introduce.php?ic=D000>).

Appendix A2

Categories covered in Taiwan's semiconductor supply chain, as classified by the SASB.

Industries classified by SASB	Sub-industries classified by SASB
Technology and communication	Semiconductor
Renewable resources and alternative energy	Hardware
Resource conversion	Electronic manufacturing services and original design manufacturing
Consumer goods	Electrical and electronic equipment
Public construction	Software and IT services
Service	Solar technology and project developers
	Industrial machinery and articles
	Multiline and specialty retailers and distributors
	Chemical
	Waste management
	Professional and business services
	Real estate services
	Apparel, Accessories & Footwear

Appendix A3

Description of variables and data sources.

Variable	Description	Data Source
<i>ESG</i>	<i>ESG</i> combined score, value = 0–100.	TEJ TEGS Sustainability Indicators database
<i>Env</i>	Environmental score, value = 0–100.	TEJ TEGS Sustainability Indicators database
<i>Soc</i>	Social score, value = 0–100.	TEJ TEGS Sustainability Indicators database
<i>Gov</i>	Governance score, value = 0–100.	TEJ TEGS Sustainability Indicators database
<i>Peer</i>	Natural logarithm of the peer average score in the upstream, midstream, or downstream sector, excluding the focal company. $Peer_{jt} = \frac{\sum_{q \in j, q \neq i} Score_{qt}}{N_{jt} - 1}$, where <i>i</i> denotes focal company, <i>j</i> denotes the upstream, midstream, or downstream sector, <i>q</i> denotes the peer company in industry stream <i>j</i> , and <i>N_{jt}</i> denotes the total number of companies in industry stream <i>j</i> in year <i>t</i> .	TEJ TEGS Sustainability Indicators database
<i>Rank</i>	A company's score ranking in subsectors classified by the SASB. $rank_{qt} = 1 - \frac{score_rank_{iqt}}{N_{qt}}$, where $rank_{qt}$ denotes company <i>i</i> 's score ranking scaled by the total number of rated companies in subsector <i>q</i> in year <i>t</i> . $score_rank_{iqt}$ is the score ranking of company <i>i</i> in SABA subsector <i>q</i> in year <i>t</i> , and <i>N_{qt}</i> is the total number of rated companies in sub-sector <i>q</i> in year <i>t</i> .	TEJ TEGS Sustainability Indicators database
<i>UP/Mid/Down</i>	Natural logarithm of the peer average score in the upstream, midstream, or downstream sectors for vertical peer spillovers.	TEJ TEGS Sustainability Indicators database

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Appendix A3 (continued)

Variable	Description	Data Source
	$Up Mid Down_{jt} = \frac{\sum_{q \in j} Score_{qt}}{N_{jt}}$, where $Up Mid Down_{jt}$ is the average score of upstream, midstream, or downstream sector j in year t , $Score_{qt}$ is the score of peer company g in industry stream j in year t , and N_{jt} is the total number of companies in industry stream j in year t .	
<i>Highsale</i>	Degree of customer transaction linkage. The net revenue of all companies is sorted in descending order by year, and the samples are divided into three equal groups by year. <i>Highsale</i> = 1 if company's net revenue is the top 1/3 of all companies and 0 otherwise.	TEJ Finance database
<i>Highcost</i>	Degree of supplier transaction linkage. The operating cost ratio of all companies is sorted in descending order by year, and the samples are divided into three equal groups by year. <i>Highcost</i> = 1 if company's operating cost ratio is the top 1/3 of all companies and 0 otherwise.	TEJ Finance database
<i>Positive</i>	Annual change in the ESG combined score and three pillar scores. <i>Positive</i> = 1 if score change >0 and 0 otherwise.	TEJ TSEG Sustainability Indicators database
<i>Negative</i>	Annual change in the ESG combined score and three pillar scores. <i>Negative</i> = 1 if score change <0 and 0 otherwise.	TEJ TSEG Sustainability Indicators database
<i>CR3</i>	$CR_n = \sum_{i=1}^n w_{it}$, where w_{it} is the weight of the total assets (net revenue) of the top n largest companies by industry stream and year.	TEJ Finance database
<i>HHI</i>	$HHI = \sum_{i=1}^n w_{it}^2$, where w_{it} is the weight of the total assets (net revenue) of each company by industry stream and year.	TEJ Finance database
<i>Lnta</i>	The natural logarithm of total assets.	TEJ Finance database
<i>ROA</i>	The ratio of EBIT to average total assets.	TEJ Finance database
<i>Debt ratio</i>	The ratio of total debt to total assets.	TEJ Finance database
<i>MBratio</i>	The market value of equity to the book value of equity.	TEJ Finance database

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